

Introduction to First-Order Logic

First-Order Logic vs. Propositional Logic

- **Propositional Logic:** Only deals with simple propositions that are either true or false.
- First-order logic (FOL), also known as **predicate logic** or **first-order predicate logic**, is a formal system used in mathematics and computer science to represent and reason about propositions that involve objects and their relationships.
- It extends propositional logic by introducing the concepts of **objects**, **quantifiers**, **variables**, and **predicates**, allowing more expressive statements about the properties of objects and their relationships.

Key Components of First-Order Logic

- 1.Constants:** These represent specific objects in the domain of discourse. For example, in a logic about people, constants could be "John" or "Mary."
- 2.Variables:** These are placeholders that can represent any object in the domain. They are typically written as x , y , z , etc.
- 3.Predicates:** These are functions that return true or false, depending on the relationship between objects. For example, $\text{Likes}(x,y)$ could be a predicate meaning "x likes y."
- 4.Functions:** These map objects to other objects within the domain. For example, $\text{fatherOf}(x)$ might return the father of x .

5. Quantifiers:

- **Universal Quantifier ($\forall$):** It expresses that a statement applies to all objects in the domain. For example, $\forall x$ means "for all x."
- **Existential Quantifier ($\exists$):** It expresses that there exists at least one object for which a statement is true. For example, $\exists x$ means "there exists an x."

6. Logical Connectives:

- **Conjunction ($\wedge$):** "And"
- **Disjunction ($\vee$):** "Or"
- **Negation ($\neg$):** "Not"
- **Implication ($\rightarrow$):** "If... then..."
- **Biconditional ($\leftrightarrow$):** "If and only if"

A BNF for FOL

S := <Sentence> ;
<Sentence> := <AtomicSentence> |
 <Sentence> <Connective> <Sentence> |
 <Quantifier> <Variable>, ... <Sentence> |
 "NOT" <Sentence> |
 "(" <Sentence> ")";
<AtomicSentence> := <Predicate> "(" <Term>, ... ")" |
 <Term> "=" <Term>;
<Term> := <Function> "(" <Term>, ... ")" |
 <Constant> |
 <Variable>;
<Connective> := "AND" | "OR" | "IMPLIES" | "EQUIVALENT";
<Quantifier> := "EXISTS" | "FORALL" ;
<Constant> := "A" | "X1" | "John" | ... ;
<Variable> := "a" | "x" | "s" | ... ;
<Predicate> := "Before" | "HasColor" | "Raining" | ... ;
<Function> := "Mother" | "LeftLegOf" | ... ;

Quantifiers

- **Universal quantification**

- $(\forall x)P(x)$ means that P holds for **all** values of x in the domain associated with that variable
- E.g., $(\forall x) \text{dolphin}(x) \Rightarrow \text{mammal}(x)$ **all dolphins are mammals**
 $(\forall x) \text{human}(x) \Rightarrow \text{mortal}(x)$ **all humans are mortal**

- **Existential quantification**

- $(\exists x)P(x)$ means that P holds for **some** value of x in the domain associated with that variable
- E.g., $(\exists x) \text{mammal}(x) \wedge \text{lays-eggs}(x)$ **there is at least one mammal that lays eggs**
 $(\exists x) \text{taller-than}(x, \text{Fred})$ **there is at least one individual who is taller than Fred**
 $(\exists x) \text{CSE-Student}(x) \wedge \text{taller-than}(x, \text{Fred})$ **there is at least one CSE student who is taller than Fred**

Quantifiers

- Universal quantifiers are often used with “implies” to form “rules”:
 $(\forall x) \text{ student}(x) \rightarrow \text{smart}(x)$ means “All students are smart”
- Universal quantification is *rarely* used to make blanket statements about every individual in the world:
 $(\forall x) \text{ student}(x) \wedge \text{smart}(x)$ means “Everyone in the world is a student and is smart”
- Existential quantifiers are usually used with “and” to specify a list of properties about an individual:
 $(\exists x) \text{ student}(x) \wedge \text{smart}(x)$ means “There is a student who is smart”
- A common mistake is to represent this English sentence as the FOL sentence:
 $(\exists x) \text{ student}(x) \rightarrow \text{smart}(x)$
 - But what happens when there is a person who is *not* a student?

Quantifier Scope

- Switching the order of universal quantifiers *does not* change the meaning:
 - $(\forall x)(\forall y)P(x,y) \leftrightarrow (\forall y)(\forall x) P(x,y)$
 - for every x and for every y , the predicate $P(x,y)$ is true.
 - for every y and for every x , the predicate $P(x,y)$ is true.
- Similarly, you can switch the order of existential quantifiers:
 - $(\exists x)(\exists y)P(x,y) \leftrightarrow (\exists y)(\exists x) P(x,y)$
 - there exists some x and there exists some y such that the predicate $P(x,y)$ is true.
 - there exists some y and there exists some x such that the predicate $P(x,y)$ is true.

- Switching the order of universals and existentials *does* change meaning:
- Everyone likes someone: $(\forall x)(\exists y) \text{ likes}(x,y)$
- for every x , there exists some y such that x likes y .
- In other words, everyone likes at least one person or thing.
- Someone is liked by everyone: $(\exists y)(\forall x) \text{ likes}(x,y)$
- This means "there exists some y such that for every x , x likes y ."
- In other words, there is a particular person or thing that everyone likes.

Connections between All and Exists

We can relate sentences involving $\forall$ and $\exists$ using **De Morgan's laws**:

$$(\forall x) \neg P(x) \leftrightarrow \neg(\exists x) P(x)$$

This can be read as: "For all x, not P(x)" is logically equivalent to "It is not the case that there exists an x such that P(x)."

$$\neg(\forall x) P \leftrightarrow (\exists x) \neg P(x)$$

It is not the case that for all x, P(x)" is logically equivalent to "There exists an x such that not P(x)

$$(\forall x) P(x) \leftrightarrow \neg (\exists x) \neg P(x)$$

For all x, P(x)" is logically equivalent to "It is not the case that there exists an x such that not P(x)

$$(\exists x) P(x) \leftrightarrow \neg(\forall x) \neg P(x)$$

There exists an x such that P(x)" is logically equivalent to "It is not the case that for all x, not P(x)

Translating English to FOL

- "All dogs are mammals."
- FOL: $\forall x(\text{Dog}(x) \rightarrow \text{Mammal}(x))$
- Explanation: For all x , if x is a dog, then x is a mammal.
- "Some students are hardworking."
- FOL: $\exists x(\text{Student}(x) \wedge \text{Hardworking}(x))$
- Explanation: There exists at least one x such that x is a student and x is hardworking.
- "John likes pizza and pasta."
- FOL: $\text{Likes}(\text{John}, \text{Pizza}) \wedge \text{Likes}(\text{John}, \text{Pasta})$
- Explanation: John likes pizza and John likes pasta.
- "No fish can fly."
- FOL: $\neg \exists x(\text{Fish}(x) \wedge \text{CanFly}(x))$
- Explanation: There does not exist any x such that x is a fish and x can fly.

Convert NL to FOL

- Consider the following story of the "lucky student":
"Anyone passing his history exams and winning the lottery is happy. But anyone who studies or is lucky can pass all his exams. John did not study but he is lucky. Anyone who is lucky wins the lottery."

Answer

- **Identify predicates:** $P(x)$: x passes history exams $W(x)$: x wins the lottery $H(x)$: x is happy $S(x)$: x studies $L(x)$: x is lucky

Now, let's formalize the given statements:

1. Anyone passing his history exams and winning the lottery is happy: $\forall x((P(x) \wedge W(x)) \rightarrow H(x))$
2. Anyone who studies or is lucky can pass all his exams: $\forall x((S(x) \vee L(x)) \rightarrow P(x))$
3. John did not study but he is lucky: $\neg S(\text{John}) \wedge L(\text{John})$
4. Anyone who is lucky wins the lottery: $\forall x(L(x) \rightarrow W(x))$

Quantified inference rules

- Universal instantiation
 - $\forall x P(x) \therefore P(A)$
- Universal generalization
 - $P(A) \wedge P(B) \dots \therefore \forall x P(x)$
- Existential instantiation
 - $\exists x P(x) \therefore P(F)$ $\leftarrow$ skolem constant **F**
- Existential generalization
 - $P(A) \therefore \exists x P(x)$

Universal instantiation

(a.k.a. universal elimination)

- If $(\forall x) P(x)$ is true, then $P(C)$ is true, where C is *any* constant in the domain of x
- Example:
$$(\forall x) \text{ eats}(\text{Ziggy}, x) \Rightarrow \text{eats}(\text{Ziggy}, \text{IceCream})$$
- The variable symbol can be replaced by any ground term, i.e., any constant symbol or function symbol applied to ground terms only

Existential instantiation

(a.k.a. existential elimination)

- From $(\exists x) P(x)$ infer $P(c)$
- Example:
 - $(\exists x) \text{ eats}(\text{Ziggy}, x) \rightarrow \text{eats}(\text{Ziggy}, \text{Stuff})$
- Note that the variable is replaced by a **brand-new constant** not occurring in this or any other sentence in the KB
- Also known as skolemization; constant is a **skolem constant**
- In other words, we don't want to accidentally draw other inferences about it by introducing the constant
- Convenient to use this to reason about the unknown object, rather than constantly manipulating the existential quantifier

Existential generalization (a.k.a. existential introduction)

- If $P(c)$ is true, then $(\exists x) P(x)$ is inferred.
- Example
$$\text{eats}(\text{Ziggy}, \text{IceCream}) \Rightarrow (\exists x) \text{eats}(\text{Ziggy}, x)$$
- All instances of the given constant symbol are replaced by the new variable symbol
- Note that the variable symbol cannot already exist anywhere in the expression

Unification

- It's the process of finding a substitution that makes two logical expressions identical. Let's dive into an explanation with examples.
- Definition: Unification is an algorithm that takes two terms and attempts to find a substitution that makes them equal. If such a substitution exists, the terms are said to be unifiable.
- Key Concepts:

Example 1: Simple Unification

- Term 1: $f(x, g(y))$
- Term 2: $f(A, g(B))$

To unify these terms:

1. The outermost functors (f) match, so we continue.
 2. For the first argument, we substitute x with A .
 3. For the second argument, the functors (g) match, so we continue.
 4. Inside g , we substitute y with B .
- The resulting unifier is $\{x/A, y/B\}$, which means:
 - x is replaced by A
 - y is replaced by B
 - After applying this unifier, both terms become: $f(A, g(B))$

Example 2: Failed Unification

- Term 1: $f(x, g(x))$ Term 2: $f(A, g(B))$
- These terms cannot be unified because:
 1. The first argument suggests x should be A .
 2. But the second argument suggests x should be B .
 3. Since $A \neq B$, there's no consistent substitution for x .